

COURSE TITLE : TOOL ENGINEERING LAB - III
COURSE CODE : 4119
COURSE CATEGORY : A
PERIODS/WEEK : 3
PERIODS/SEMESTER : 45
CREDITS : 2

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Pneumatics Lab	15
2	Bench Work	18
3	Practice on Jig Boring Machine	12
	Total	45

COURSE OUT COMES

On completion of course, student should be able to:

1. Understand pneumatic circuits of various machines.
2. Understand the working, connections of single acting, double acting air cylinders.
3. Understand the working, connections of direction control valves..
4. Practice the blue matching of mating surfaces such as core and cavity of mould
5. Practice various operations on jig boring machines.

CONTENT OUTLINE

1. Pneumatics Lab

- i) Simple Circuit Drawing
- ii) Circuit of Single acting cylinder
- iii) Circuit of double acting cylinder
- iv) Circuit using different types of direction control valves

2. Bench work

- i) Blue matching of core and cavity of moulds

3. Practice on Jig Boring Machines

(Work related to the manufacture of press tool components.)